

App Development Project Report

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1. Student Details

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About Me: I am a Student at IIT Madras BS Degree program with a deep interest in data analytics and machine learning. I enjoy studying about various latest technologies and upgrading myself in these fields.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and develop a centralized web-based platform that streamlines the campus placement process by connecting students, companies, and administrators. The system should allow companies to post job drives, students to apply, and admins to monitor and manage the entire workflow..

Approach:

The application was developed using Flask as the backend framework with a structured modular design. The system consists of three user roles:

- **Admin:** Manages students, companies, drives, and applications
- **Company:** Posts job drives and reviews applicants
- **Student:** Views drives and applies to them

The backend handles authentication, database operations, and routing, while the frontend uses Jinja templates and CSS for dynamic rendering and UI consistency.

3. AI/LLM Declaration

I used ChatGPT (GPT-5) to assist with:

- Improving UI layout and structure
- Enhancing HTML/CSS for better user experience
- Documentation formatting

The extent of AI usage is approximately **15-17%**, mainly for guidance in the UI layout. All logic implementation, integration, and testing were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend web framework
SQLAlchemy	Object Relational Mapper for database handling
Jinja2	Dynamic HTML templating
Bootstrap 5	Frontend UI design
Flask session	Authentication handling
SQLite	Database

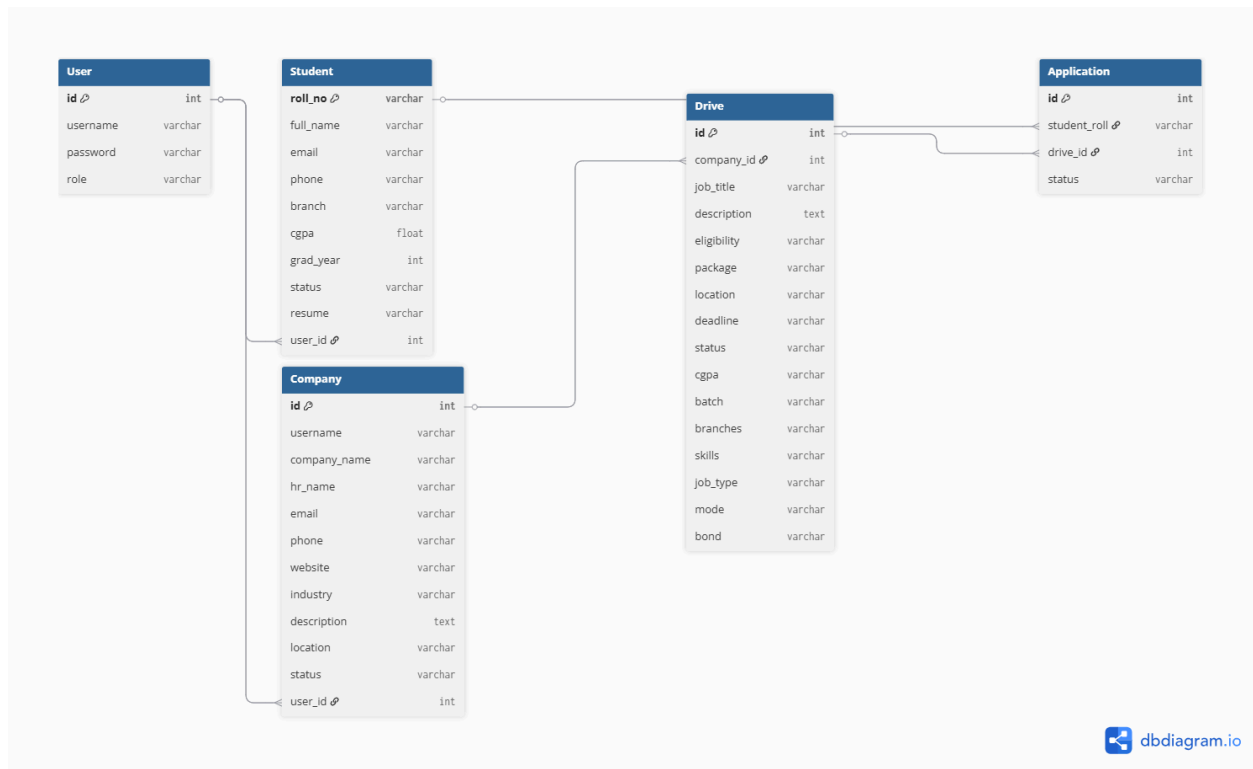
5. Database Schema / ER Diagram

Tables:

1. User: (id (Primary Key), username (Unique), password, role (admin/company/student))
2. Student: (roll_no (Primary Key), full_name, email, phone, branch, cgpa, grad_year, status (pending/approved/rejected/blacklisted), resume)
3. Company: (id (Primary Key), username (Unique), company_name, hr_name, email, phone, website, industry, description, location, status (pending/approved/rejected))
4. Drive: (id (Primary Key), company_id (Foreign Key), job_title, description, eligibility, package, location, deadline, status (pending/approved/rejected), cgpa, batch, branches, skills, job_type, mode, bond)
5. Application: (id (Primary Key), student_roll (Foreign Key), drive_id (Foreign Key), status (pending/applied/shortlisted/selected/rejected))

Relationships:

- **User** → **Student** : One-to-One
- **User** → **Company** : One-to-One
- **Company** → **Drive** : One-to-Many
- **Student** → **Application** : One-to-Many
- **Drive** → **Application** : One-to-Many



6. API Resource Endpoints

Endpoint	Method	Description
/login	GET, POST	User login (Admin / Student / Company)

<code>/logout</code>	GET	Logout user session
<code>/register/student</code>	GET, POST	Student registration
<code>/register/company</code>	GET, POST	Company registration
<code>/student/dashboard</code>	GET	Student dashboard
<code>/student/drives</code>	GET	View available drives
<code>/student/history</code>	GET	View application history
<code>/company/dashboard</code>	GET	Company dashboard
<code>/admin/dashboard</code>	GET	Admin dashboard
<code>/admin/students</code>	GET	View all students
<code>/admin/companies</code>	GET	View all companies
<code>/admin/drives</code>	GET	View all drives
<code>/admin/applications</code>	GET	View all applications

7. Architecture and Features

Architecture Overview:

- **app.py** – main Flask application entry point
- **models.py** – database models using SQLAlchemy
- **/templates** – HTML templates
- **/static** – CSS and UI styling

Implemented Features:

- Role-based login (Admin / Company / Student)
- Admin dashboard with analytics
- Company registration & approval
- Drive creation and management
- Student dashboard with drive listings
- Apply to drives functionality
- Application tracking system
- Status updates (shortlisted / selected / rejected)
- Clean UI with sidebar navigation

Additional Features:

- Responsive UI design
- Dashboard cards for quick insights
- Application history with status badges
- Profile management for students

8. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1sEGkU2-vJLVq5xxbqdiKU-fjenrVTK8W/view?usp=sharing>